

Eccentric training: The key for a stronger, more resilient athlete?

Jan Wilke and Håkan Alfredson

In sports and exercise, concentric contraction is often regarded as the superhero of muscle work: basketball and volleyball players are interested in the achieved jump height, sprinters are concerned about their legs' rate of force development, and injured football (soccer) players are cleared for return to play once having restored their pre-injury strength levels, alongside other factors. Despite the unquestionable relevance of these aspects, concentric contraction has a fascinating twin, which rarely enters the spotlight. Eccentric contraction plays a vital role in both injury prevention and sports performance. This chapter provides a comprehensive and practice-oriented review of the basic mechanisms, related training methods and fields of application in sports and exercise.

Basic physiology of eccentric contraction

Eccentric contraction (see Figure 26.1 for an overview of contraction types) is what makes life feel smooth and easy. When descending a flight of stairs or sitting down on a chair, we do not let ourselves fall down passively, which could be painful. Instead, prior to making contact with the ground or an object, we activate our internal “brakes” using this special type of muscle work: in the case of sitting down, the calf, quadriceps and gluteus maximus muscles are stretched while simultaneously developing force in order to decelerate the downward movement. As the

term “decelerate” indicates, it is a hallmark of eccentric muscle work that the amount of force lengthening the muscle exceeds the force output it produces. Using the feedback of the joint proprioceptors (muscle spindle, Golgi tendon-like organ, Ruffini and Pacinian corpuscles), the most crucial task of the central nervous system is to meticulously adapt the required contraction level according to the requirements of each situation in daily life, as well as during exercise. A better understanding of the role of eccentric contraction as a shock-absorbing servo-mechanism is therefore highly beneficial for health professionals and performance coaches.

In the sarcomere, the stretch occurring during eccentric contractions leads to rapid detachments and re-attachments of the cross-bridges between the actin and myosin filaments. On a more macroscopic level, the superordinate functional units (sarcomere, myofibrils, myofibers and the whole muscle) are successively elongated. Experiments in the 1970s revealed an interesting phenomenon: during a contracted stretch, considerable force increases can be registered (Edman et al., 1976). If tissue lengthening ends but muscle activation is maintained (e.g. at the end point of an eccentric movement), one part of this extra force is lost. However, the other part, making the contraction stronger than a normal isometric contraction at the same muscle length, remains until the muscle is relaxed or returned to its resting length. Extensive research has been

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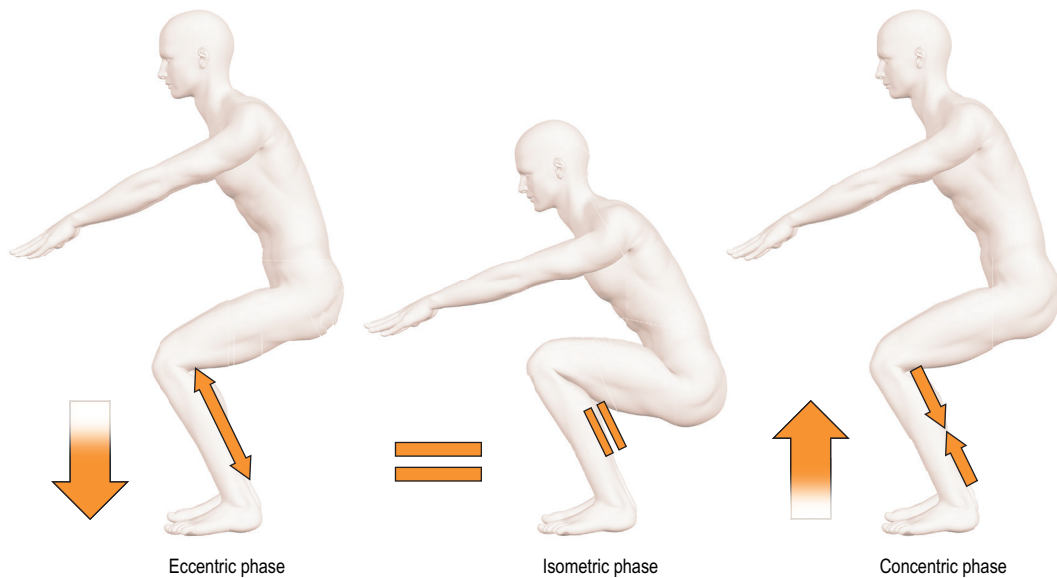


Figure 26.1

Schematic illustration of different contraction types. When eccentrically descending into a deep squat position (left), the gastrocnemius is lengthened. If the deep position is held isometrically for a few seconds (middle), the muscle length remains constant. When returning to the upright position, the gastrocnemius shortens due to the concentric contraction (right).

performed to identify the origin of this effect, which is called residual force enhancement. Although still not understood fully, it is assumed that the elastic properties of the spring-like titin filaments play a key role. Particularly during an active stretch or an eccentric contraction, a specific part of the titin (the PEVK region) is stiffened and stores kinetic energy. The resulting recoil forces pulling the muscle towards its resting length could explain a major part of the increased muscle force (Hessel et al., 2017). Although there is a paucity of research concerning this hypothesis, it can be assumed that the deep fascia also contributes to this rebound effect as it is stretched during the eccentric movement phase. In summary, the observation that eccentric movements allow the handling of

higher weights is attributable to two factors, (1) the nature of this contraction type, which does not aim to accelerate but to decelerate objects, and (2) the residual force enhancement combined with elastic energy storage.

Forms of eccentric training

The most traditional form of eccentric training (ET) is the performance of body weight exercises with a long eccentric contraction phase. Typical examples include lowering the center of mass in squats and push-ups as slowly as possible. For some muscles and joints, and if training progress is made, external weights may be required to achieve sufficient eccentric stimulation. In addition to strength training machines (which are not recommended in most cases because of

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the limited co-ordination challenge), barbells, dumbbells and kettlebells are well-suited for this purpose. As mentioned above, the maximal training weight is limited by concentric force. Therefore, if weights exceeding the latter are chosen, a training partner can assist during the muscle-shortening phase while not providing assistance in the heavy-loaded eccentric part of the movement. The possibility of imposing higher mechanical stresses to the targeted tissues certainly represents an advantage of this method. However, in real life, eccentric movement is almost always coupled with concentric activity, meaning that exercises with concentric support are less functional. In addition, when seen from a fascial perspective, the effective use of elastic energy stored in the intra- and extra-muscular connective tissue is not trained with this technique. In recent years, specific devices have been designed to optimize muscle loading during ET. Flywheel training is an increasingly popular, instrument-assisted form of isoinertial ET. Connected via a strap, the training individual moves the flywheel with a concentric contraction. In the subsequent eccentric movement phase, resisting the pull of the strap, the rotation of the wheel needs to be decelerated.

Morphological and mechanical adaptations of ET

When performed over a period of several weeks, ET elicits substantial muscle hypertrophy. While it is similarly effective as concentric exercise in this regard, there is an interesting region-specificity: following ET, distal muscle size increases more strongly as opposed to concentric training, where thickness changes predominantly affect the mid-portion of the muscle. Another significant adaptation of ET is a larger increase in type IIx muscle fiber cross-sectional

area (Douglas et al., 2017). The fast-type glycolytic fibers are crucial for rapid force development and hence, all athletes who require explosive movement may consider incorporating ET into their exercise routines.

ET can significantly affect the morphological and material properties of the tendon (Chapter 5), inducing increases in the cross-sectional area, stiffness and Young's modulus (Bohm et al., 2015). It further stimulates collagen synthesis, which may explain the observations described above (Kjaer et al., 2009). While all these adaptations are beneficial for athletes and clients, it has to be reiterated that the effect magnitude is not different between ET and concentric training. It is assumed that the training intensity (high loading) is more decisive than the contraction type (Kjaer et al., 2009; Bohm et al., 2015). However, as ET allows the use of higher weights, it may still be the more appropriate choice if sufficient time for progressive load increases is available.

In addition to triggering adaptations in the muscle and the tendon, ET leads to marked changes in the intra- and extra-muscular connective tissue. On days following heavy eccentric exercise until fatigue, a substantial thickening of the deep fascia can be detected using high-resolution ultrasound imaging (Nosaka and Lau, 2018). The results from other studies also suggest a profound mechanical impact on the muscle-associated soft tissue. Raastad et al. (2010) required their participants to perform 300 eccentric contractions of the knee extensors. After the training session, the authors observed increases in tenascin-C and a propeptide of procollagen III. These reflect both high extracellular matrix loading and remodeling. When compared with concentric exercise, ET provides a stronger

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stimulus for the synthesis of intramuscular collagen type I, which may lead to increased connective tissue cross-linking (Heinemeier et al., 2007). While an initial bout of eccentric exercise creates a strong tissue response, repeated training without progressive load increase does not seem to evoke the same adaptations. Mackey et al. (2011) examined the tensascin-C concentration 48 h after both an acute and an identical second electrical eccentric exercise bout (performed 30 days after the first session). In contrast to the initial bout, the repetition did not alter the expression of the protein. Most likely, a protective remodeling process rendered the connective tissue more resilient. This hypothesis is supported by the fact that the thickening response of the deep fascia (as described above), occurring after an unaccustomed bout of eccentric exercise, does not manifest after an identical repeated training session (Nosaka and Lau, 2018).

Effects of ET on performance

Several systematic reviews demonstrate that ET substantially increases absolute strength (Roig et al., 2009). With regard to the different methods, flywheel training represents the most effective type of exercise (Maroto-Izquierdo et al., 2017). A meta-analysis of 15 studies showed similar gains in concentric and eccentric exercise (Roig et al., 2009). However, when exclusively analyzing studies that used higher intensities in the eccentric exercise group than in the concentric exercise group (which is realistic because the former allows the use of higher loads), the increases in total strength were higher for ET. This implies that, as long as developing maximal strength represents the main objective, the highest possible intensity needs to be chosen if ET is used instead of concentric exercise. Not surprisingly,

in addition to maximal concentric strength, ET is more effective in developing maximal eccentric strength. Notwithstanding, this specificity is of utmost importance if an athlete or client needs to be prepared for the mechanical stresses associated with rapid deceleration.

While several studies have examined the effect of ET on maximal strength, there are less data regarding vertical jump height and sprinting speed, which are extremely important in several sports. However, from the evidence available it can be inferred that eccentric exercise, particularly when performed at high movement speeds, seems to improve both parameters more effectively than concentric exercise: in their systematic review, Douglas et al. (2017) reported stronger improvements of stretch-shortening cycle movement following eccentrics compared with traditional concentric exercise. Similar findings were reported by Maroto-Izquierdo et al. (2017), who additionally revealed the superiority of flywheel training over traditional ET with free weights or machines.

Besides improving strength and explosive movement capacities, ET of the lower limb has been demonstrated to enhance flexibility in the ankle, knee and hip joint. The magnitude of this effect is comparable with that of static stretching interventions (Sullivan et al., 2012). Although different factors are discussed, the ROM increases appear to result from the formation of new sarcomeres. Sarcomerogenesis has been confirmed in animal studies and represents an adaptation optimizing the length-tension relationship of the muscle (Sullivan et al., 2012).

ET and musculoskeletal disorders

While ET can increase strength in the long term, a single eccentric contraction does not make you run substantially faster or hit the ball harder.

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However, its impact in sports cannot be over-rated in the short term either. Rapid deceleration, which is based on eccentrics, is key for quick changes of direction and, even more importantly, effective injury prevention. Up to 77% of all non-contact traumas occurring in game sports are due to failed jump landings (Woods et al., 2003). One important strategy to reduce this dramatically high incidence is teaching proper landing techniques based on eccentric contraction (Sinsurin et al., 2016). Imagine the ground contact after a depth jump performed with either extended or flexed knees. With the former, there is no substantial involvement of the leg muscles in the dampening of the impact. By contrast, with dynamic knee flexion, the landing is softened due to the eccentric contraction (from extended to flexed knees) of the glutes and the quadriceps muscle.

Besides improving landing technique (which involves a better acute use of eccentrics within a specific movement), regular eccentric exercise has been shown to be an effective strategy to prevent (myofascial) muscle injuries. Hamstring strains are the most frequent disorder in sports, particularly in football/soccer (Edouard et al., 2015; Ernlund and Vieira, 2017).

ET, when performed regularly, can reduce the injury risk by as much as 65% (Goode et al., 2015). The signature exercise to be used here is Nordic Hamstring Lowering (Figure 26.2).

ET is frequently performed in the rehabilitation of musculoskeletal disorders. Lateral epicondylitis mainly affects tennis players and causes pain that may radiate to the hand and shoulder. A multimodal therapy with ET improves pain, function and grip strength more effectively than interventions without ET (Cullinane et al., 2014). A simple exercise can be performed with small dumbbells (2–5 kg) or a water bottle. Positioning the lower arm on a table, the wrist is extended (with support of the free hand if needed) and then slowly lowered into flexion.

Tendinopathy represents another pathology that is frequently treated by means of ET. An extended description with an exemplary loading protocol for the Achilles tendon is provided at the end of this chapter. One recent meta-analysis concludes that there are still insufficient high-quality data to conclusively judge the effectivity of ET in Achilles tendinopathy compared with other treatments, although it is assumed that ET is more effective than normal physiotherapy (Murphy et al., 2019). In patients



Figure 26.2

The Nordic Hamstring exercise is an excellent method to increase eccentric strength of the dorsal thigh muscles. The movement is initiated in an upright position (left). With the feet fixed (e.g. by a coach) and the hip extended, the body is slowly lowered towards the ground (middle). Only if the movement can no longer be controlled, the arms are used to cushion the fall (right).

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with Patellar tendinopathy, ET performed in conjunction with or without stretching and core stabilization exercises leads to a mean improvement of 61% in subjectively rated symptoms (Everhardt et al., 2017). A suitable exercise to intensify the eccentric loading of the Patellar tendon is the unilateral or bilateral squat using a decline board (Figure 26.3).

ET and the prevention of tendon disorders

In view of the beneficial effects of ET in the therapy of tendinopathy, it seems reasonable to use the method as a prophylactic intervention in asymptomatic individuals. However, available data from prospective studies do not indicate an injury-preventive effect. Based on current evidence, therefore, ET cannot be recommended with regard to primary prevention (Peters et al., 2016).

Important aspects to consider in ET

- ET imposes substantial stresses on the locomotor system. Coaches and therapists should therefore ensure sufficient time for gradual tissue adaptation. When designing training interventions, start with slow concentric and isometric movements and then progress to eccentric movements. Similarly, begin with body weight only, step by step increasing external loads.
- If large strength gains or muscle hypertrophy are desired, controlled single-joint exercises may be an adequate choice. However, from a functional point of view, the additive use of multi-joint movements is essential to transfer and exploit triggered adaptations in the target sport.
- If rapid eccentric movements are intended to be trained, the use of heavy barbells or



Figure 26.3

A decline board can be used during a conventional squat to increase eccentric loading of the quadriceps femoris muscle. Keeping the body upright, the knee is progressively flexed as far as possible (left). To increase intensity, besides performing the exercise on one leg, a backpack with weights can be worn (right).

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dumbbells may not be optimal. In this case, elastic bands can provide a workable solution.

- While athletes from multiple sports benefit from lower extremity eccentric exercises (e.g. Nordic Hamstrings, Decline Squats/Lunges), throwers can also effectively develop eccentric strength. For deceleration throws, use small balls of varying weights (start light), instructing the athletes to catch them dynamically, following the direction of the ball. Additionally, traditional exercises (e.g. bench press) can be used to increase eccentric strength.

Conclusion

Performing eccentric exercises is a highly effective method to elicit both functional and structural changes in muscles, tendons and the extra-muscular connective tissue. Besides improving performance, it seems to be of value in the prevention and treatment of musculoskeletal disorders and, therefore, the use of ET should always be considered when designing long-term training regimes. If athletes suffer from tendinopathy, ET represents a gold standard method to alleviate the symptoms with minimal use of medication.

Clinical advice from the expert *Håkan Alfredson*

Basic rationale for the use of eccentric exercise in tendinopathy

Curwin and Stanish (1986) were the first to raise the theory of eccentric exercises being included in the treatment of painful tendinopathy. They proposed that pain-free eccentric exercises, performed with gradually increased speed, should be used on patients with mid-portion Achilles and patellar tendinopathy. However, no patient material on a specific diagnosis was presented in a scientific study. After trying eccentric exercises on my own chronically painful

Achilles mid-portion, with a modified version including painful heavy eccentric loading and multiple repetitions performed at slow speed, the successful result was the beginning for research projects evaluating large patient cohorts, with the diagnosis of mid-portion Achilles tendinopathy/tendinosis verified by means of ultrasound.

Scientific evaluation of the method

Our treatment was initially tried on patients on our waiting list for, at that time, traditional open tendon surgery (including tenotomy, excision of tendinosis tissue, and a 4-6 months postoperative rehabilitation). In the pilot study, all 15 included patients were satisfied with the treatment, and did not want to have surgery (Alfredson et al., 1998). After 1 year, one patient had regained pain symptoms and this patient was surgically treated. The other 14 patients were still active in Achilles tendon loading activities. Following this pilot study, we performed multiple studies on the effects of ET, including a randomized study and follow-up studies with large sample sizes (Mafi et al., 2001; Fahlström et al., 2003).

An interesting observation is that in successfully treated patients, the tendon thickness is decreased and the structure is sonographically normalized (Öhberg et al., 2004). Consequently, it seems that eccentric calf muscle training has a potential to remodel, and possibly also induce, a regeneration of the previously so-called degenerative tendon.

We have learned that, overall, the results of heavy-load ET are beneficial in about 80% of cases. However, it seems that high-level athletes, especially spike-shoe runners and jumpers, have less positive results with this treatment. Also, it is important that the patients have a correct diagnosis before this treatment is initiated. If eccentric exercises are used on a patient with a partial tendon rupture, there is a high risk of full-thickness rupture, a condition that is very difficult to treat, and may render the patient with permanently reduced calf muscle strength and function.

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Observations during the last 7–8 years have shown that there is a subgroup of patients with mid-portion Achilles tendinopathy that exhibit a plantaris tendon involvement (Alfredson, 2011). The plantaris tendon can be considered as a remnant from earlier human development stages when we were moving on four legs (Spang et al, 2016). This thin tendon, which occasionally develops tendinopathies, thickening and widening at the medial side of the Achilles, has been shown to have multiple different positions in relation to the Achilles tendon. In rare cases it is even invaginated into the medial side of the Achilles. Patients with plantaris involvement frequently have a poor response to ET, and sometimes even a worsening of the pain condition occurs. If plantaris involvement is suspected, coaches and therapists should ask for the exact pain location, as patients typically report complaints at the medial side of the Achilles. The soft tissues in between the plantaris and Achilles tendons have been shown to be richly innervated, alongside the plantaris itself, and, in contrast to the Achilles, have been shown to frequently contain multiple sensory nerves. The plantaris tendon has been shown to mechanically interfere with the medial side of the Achilles tendon, and often requires surgical removal.

Physiology

The mechanisms behind good clinical results in a chronically painful tendon have yet not been fully clarified. There are multiple theories, such as increased type 1 collagen formation, increased calf muscle strength, and altered shearing forces within the tendon. However, patients often experience pain relief within 2–4 weeks from when they begin the exercises, although the time needed for collagen formation and muscle strength increases is much longer. A recent theory suggests that the effects are due to shearing forces between the ventral side of the Achilles tendon and the Kager fat tissue. This shearing mechanism is

demonstrated by movement from plantar flexion to dorsiflexion in the tendon, when the ventral fat tissue moves in opposite directions. The theory fits with observations from biopsies, in that the nerves responsible for the pain in this condition are located in the fat tissues close to the ventral side of the Achilles. This has been tested using local anesthetic injections (temporary total pain relief) and surgical scraping procedures (permanent pain relief) outside the ventral side of the tendon.

Eccentric calf muscle training in Achilles tendinopathy

The following tips may be helpful when performing eccentric exercise to alleviate tendon pain:

- Use stable shoes to prevent pain in the plantar fascia, midfoot or toe joints.
- If knee arthrosis or low-back pain is present it may not be possible to do these exercises.
- Start by using the body weight only. If no tendon pain manifests during the exercise, increase the load by using weights in a back pack. Gradually increase the load to reach a new level of “painful eccentric loading”.
- The easiest way to perform the exercises is by standing on a staircase, using the bannisters for stability. If high loads are required in the future, a standing weight machine can be used.
- Start from a tip-toe position with all the body weight on the injured side, then slowly lower the heel beneath the step until you reach bottom position (Figure 26.4). Next, support yourself to return to the start position by using the other asymptomatic leg. In bilateral Achilles tendinopathy use both legs.
- Perform 3 x 15 repetitions with straight knees and 3 x 15 repetitions with the knees bent. Carry this out two times each day for a total of 12 weeks.

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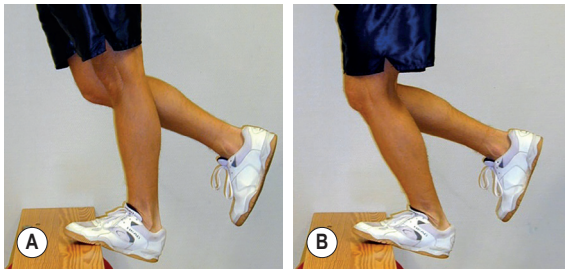


Figure 26.4

Eccentric calf muscle training with (A) straight knee and (B) bent knee.

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